34.7: Humans are mammals that have a large brain and bipedal locomotion

In our tour of Earth's biodiversity, we turn now to our own species, *Homo sapiens*, which is about 200,000 years old. When you consider that life has existed on Earth for at least 3.5 billion years, we are clearly evolutionary newcomers.

CheckPoint

Making Inferences from Phylogenetic Trees

Check your understanding before you move on, with the CheckPoint: Making Inferences from Phylogenetic Trees.

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Derived Characters of Humans

Many characters distinguish humans from other apes. Most obviously, humans stand upright and are bipedal (walk on two legs). Humans have a much larger brain and are capable of language, symbolic thought, artistic expression, and the manufacture and use of complex tools. Humans also have reduced jawbones and jaw muscles, along with a shorter digestive tract.

At the molecular level, the list of derived characters of humans is growing as scientists compare the human and chimpanzee genomes. Although the two genomes are 99% identical, a difference of 1% can represent many changes in a genome with 3 billion base pairs. Furthermore, changes in a small number of genes can have large effects. This point was highlighted by recent results showing that humans and chimpanzees differ in the expression of 19 regulatory genes. These genes turn other genes on and off and hence may account for many differences between humans and chimpanzees.

Bear in mind that such genomic differences—and whatever derived phenotypic traits they encode—separate humans from other *living* apes. But many of these new characters first emerged in our ancestors, long before our own species appeared. We will consider some of these ancestors to see how these characters originated.

HHMI BioInteractive Video

Great Transitions: The Origin of Humans

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The Earliest Hominins

The study of human origins is known as **paleoanthropology**. Paleoanthropologists have unearthed fossils of approximately 25 extinct species that are more closely related to humans than to chimpanzees. These species are known as **hominins(Figure 34.47)**. Since 1994, fossils of four hominin species dating to more than 4 million years ago have been discovered. The oldest of these hominins, *Sahelanthropus tchadensis*, lived about 6.5 million years ago.

Figure 34.47

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A timeline for selected hominin species.
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A timeline for selected hominin species.

Most of the fossils illustrated here come from sites in eastern and southern Africa. Note that at most times in hominin history, two or more hominin species were contemporaries.

A table is given:

Time Start (millions of years ago)	Time Stop (millions of years ago)	Hominin species	Fossil Description
Just over 6.5	Just under 6.5	Sahelanthropus tchadensis	Flat skull top
Just under 6.0	Just over 6.0	Orrorin tugenensis	Bone fragments only
About 5	About 4.4	Ardipithecus ramidus	Rounder skull top, fragment
About 4.2	About 3.9	Australopithecus anamensis	Jaw only
4	3	Australopithecus afarensis	Fairly complete upper portion of skull,
About 3.6	About 3.4	Kenyanthropus platyops	Upper skull fragment

3	About 2.4	Australopithecus africanus	Fairly complete upper skull, rounder top,
2.5	Just under 2.5	Australopithecus garhi	Upper jaw region only.
About 2.8	1.5	Homo habilis	Larger posterior skull region, flat face
About 2.7	About 1.8	Homo rudolfensis	Larger posterior skull region, flat face
About 2.6	About 1.4	Paranthropus boisei	Shorter face, protrusion over eye
2	1.5	Paranthropus robustus	Fragment only
2	1	Homo ergaster	Flat face, round posterior skull region,
1.8	About 0.3 - uncertain	Homo erectus	Flat face, round posterior skull region,
About 0.3	About 0.1	Homo neanderthalensis	Flat face, large, round posterior skull
About 0.2	Present	Homo sapiens	Round, domed skill, short, flat face

Sahelanthropus and other early hominins shared some of the derived characters of humans. For example, they had reduced canine teeth, and some fossils suggest that they had relatively flat faces. They also show signs of having been more upright and bipedal than other apes. One clue to their upright stance can be found in the foramen magnum, the hole at the base of the skull through which the spinal cord passes. In chimpanzees, the foramen magnum is relatively far back on the skull, while in early hominins (and in humans), it is located underneath the skull. This position allows us to hold our head directly over our body, as early hominins apparently did as well. The pelvis, leg bones, and feet of the 4.4-million-year-old *Ardipithecus ramidus* also suggest that early hominins were increasingly bipedal **(Figure 34.48)**. (We will return to the subject of bipedalism later in the chapter.)

Figure 34.48

"Ardi," a 4.4-million-year-old fossil of *Ardipithecus ramidus*.

A skeleton is missing some components, but its jaw, eye socket, arm and finger bones, as well as leg and toe bones are present and resemble a human skeleton.

Note that the characters that distinguish humans from other living apes did not all evolve in tight unison. While early hominins were showing signs of bipedalism, their brains remained small—about 300–450 cm3 in volume, compared with an average of 1,300 cm3 for *Homo sapiens*. The earliest hominins were also small overall. *A. ramidus*, for example, is estimated to have been about 1.2 m tall, with relatively large teeth and a jaw that projected beyond the upper part of the face. Humans, in contrast, average about 1.7 m in height and have a relatively flat face; compare your own face with that of the chimpanzees in Figure 34.46d.

It's important to avoid two common misconceptions about early hominins. One is to think of them either as chimpanzees or as having evolved from chimpanzees. Chimpanzees represent the tip of a separate branch of evolution, and they acquired derived characters of their own after they diverged from their common ancestor with humans.

Another misconception is to think of human evolution as a ladder leading directly from an ancestral ape to *Homo sapiens*. This error is often illustrated as a parade of fossil species that become progressively more like ourselves as they march across the page. If human evolution is a parade, it is a very disorderly one, with many groups breaking away to wander other evolutionary paths. At times, several hominin species coexisted. These species often differed in skull shape, body size, and diet (as inferred from their teeth). Ultimately, all but one lineage—the one that gave rise to *Homo sapiens*—ended in extinction. Overall, considering the characteristics of all hominins that lived over the past 6.5 million years, *H. sapiens* appears not as the end result of a straight evolutionary path, but rather as the only surviving member of a highly branched evolutionary tree.

Australopiths

The fossil record indicates that hominin diversity increased dramatically between 4 and 2 million years ago. Many of the hominins from this period are collectively called australopiths. Their phylogeny remains unresolved on many points, but as a group, they are almost certainly paraphyletic. The earliest member of the group, *Australopithecus anamensis*, lived 4.2–3.9 million years ago, close in time to older hominins such as *Ardipithecus ramidus*.

Australopiths got their name from the 1924 discovery in South Africa of *Australopithecus africanus* ("southern ape of Africa"), which lived between 3 and 2.4 million years ago. With the discovery of more fossils, it became clear that *A. africanus* walked fully erect (was bipedal) and had human-like hands and teeth. However, its brain was only about one-third the size of the brain of a present-day human.

In 1974, in the Afar region of Ethiopia, scientists discovered a 3.2-million-year-old *Australopithecus* skeleton that was 40% complete. “Lucy,” as the fossil was named, was short—only about 1 m tall. Lucy and similar fossils have been given the species name *Australopithecus afarensis* (for the Afar region). Fossil evidence shows that *A. afarensis* existed as a species for at least 1 million years.

At the risk of oversimplifying, we could say that *A. afarensis* had fewer of the derived characters of humans above the neck than below. Lucy’s brain was the size of a softball, a size similar to that expected for a chimpanzee of Lucy’s body size. *A. afarensis* skulls also have a long lower jaw. Skeletons of *A. afarensis* suggest that these hominins were capable of arboreal locomotion, with arms that were relatively long in proportion to body size (compared to the proportions in humans). However, fragments of pelvic and skull bones indicate that *A. afarensis* walked on two legs. Fossilized footprints in Laetoli, Tanzania, corroborate the skeletal evidence that hominins living at the time of *A. afarensis* were bipedal **(Figure 34.49)**.

Figure 34.49

Evidence that hominins walked upright 3.5 million years ago.

Evidence that hominins walked upright 3.5 million years ago.

Part a. The Laetoli footprints, more than 3.5 million years old, confirm that upright posture evolved quite early in hominin history.

Part b. Artist’s reconstruction of A. afarensis, a hominin alive at the time of the Laetoli footprints, shows a cross between apes and modern humans, with a protruding face, hairy body, and upright posture.

Another lineage of australopiths consisted of the “robust” australopiths. These hominins, which included species such as *Paranthropus boisei*, had sturdy skulls with powerful jaws and large teeth, adapted for grinding and chewing hard, tough foods. They contrast with the “gracile” (slender) australopiths, including *A. afarensis* and *A. africanus*, which had lighter feeding equipment adapted for softer foods.

Combining evidence from the earliest hominins with the much richer fossil record of later australopiths makes it possible to formulate hypotheses about significant trends in hominin evolution. In the **Scientific Skills Exercise**, you’ll examine one such trend: how hominin brain volume has changed over time. Here we’ll consider two other trends: the emergence of bipedalism and tool use.

Scientific Skills Exercise

Determining the Equation of a Regression Line

AP Science Practices 4, 5

How Has Brain Volume Changed Over Time in the Hominin Lineage? The hominin taxon includes *Homo sapiens* and about 20 extinct species that are thought to represent early relatives of humans. Researchers have found that the brain volume of the earliest hominins ranged between 300 and 450cm3, similar to the brain volume of chimpanzees. The brain volumes of modern humans range between 1,200 and 1,800cm3. In this exercise, you'll examine how mean brain volume changed over time and across various hominin species.

A skull.

How the Study Was Done In this table, xi is the mean age of each hominin species, and yi is the mean brain volume cm3. Ages with negative values represent millions of years before the present (which has an age of 0.0).

A table lists hominin species, mean age, and mean brain volume.

Data from *Dean Falk, Florida State University, 2013.*

The table is as follows:

Hominin species	Mean age in millions of years; x sub i	x sub i minus x bar	Mean brain volume in cubic centimeters; y sub i	y sub y bar	left parenthesis, x sub i minus x bar, right parenthesis, times, left parenthesis, y sub i minus y bar, right parenthesis
Ardipithecus ramidus	minus 4.4	blank	325	blank	blank
Australopithecus afarensis	minus 3.4	blank	375	blank	blank
Homo habilis	minus 1.9	blank	550	blank	blank
Homo ergaster	minus 1.6	blank	850	blank	blank
Homo erectus	minus 1.2	blank	1000	blank	blank
Homo heidelbergensis	minus 0.5	blank	1200	blank	blank

Homo neanderthalensis	minus 0.1	blank	1400	blank	blank
Homo sapiens	0	blank	1350	blank	blank

INTERPRET THE DATA

How did the brain volume of hominin species change over time? In particular, is there a linear (straight-line) relationship between brain volume and time?

To find out, we'll perform a linear regression, a technique for determining the equation for the straight line that provides a "best fit" to a set of data. Recall that the equation for a straight line between two variables, x and y, is

$y=mx+b$

In this equation, m represents the slope of the line, while b represents the y-intercept (the point at which the straight line crosses the y-axis). When $m<0$, the line has a negative slope, indicating that the values of y become *smaller* as values of x become *larger*. When $m>0$, the line has a positive slope, meaning that the values of y become larger as values of x become larger. When $m=0$, y has a constant value (b).

The correlation coefficient, r, can be used to calculate the values of m and b in a linear regression:

$m=rsysx \text{ and } b=\bar{y}-m\bar{x}$.

In these equations, sx and sy are the standard deviations of variables x and y, respectively, while $\bar{x}$ and $\bar{y}$ are the means of those two variables. (See the Scientific Skills Exercise for Chapter 32 for more information about the correlation coefficient, mean, and standard deviation.)

- Calculate the means ($\bar{x}$ and $\bar{y}$) from the n=8 data points in the table. Next, fill in the $(xi-\bar{x})$ and $(yi-\bar{y})$ columns in the data table, and use those results to calculate the standard deviations sx and sy.
- As described in the Scientific Skills Exercise for Chapter 32, the formula for a correlation coefficient is

$r=\sum(xi-\bar{x})(yi-\bar{y})(n-1)(sxsy)$

- Fill in the column in the data table for the product $(xi-\bar{x})\times(yi-\bar{y})$. Use these values and the standard deviations calculated in question 1 to calculate the correlation coefficient r between the brain volume of hominin species (y) and the ages of those species (x).

- Based on the value of r that you calculated in question 2, describe in words the correlation between mean brain volume of hominin species and the mean age of the species.
-
    - Use your calculated value of r to calculate the slope (m) and the y-intercept (b) of a regression line for this data set.
- Graph the regression line for the mean brain volume of hominin species versus the mean age of the species. Be careful to select and label your axes correctly.
- Plot the data from the table on the same graph that shows the regression line. Does the regression line appear to provide a reasonable fit to the data?
- The equation for a regression line can be used to calculate the value of y expected for any particular value of x. For example, suppose that a linear regression indicated that m=2 and b=4. In this case, when x=5, we expect that y=2x+4=(2×5)+4=14. Based on the values of m and b that you determined in question 4, use this approach to determine the expected mean brain volume for a hominin that lived 4 million years ago (that is, x=−4).
- The slope of a line can be defined as m=y2−y1x2−x1, where (x1,y1) and (x2,y2) are the coordinates of two points on the line. As such, the slope represents the ratio of the rise of a line (how much the line rises vertically) to the run of the line (how much the line changes horizontally). Use the definition of the slope to estimate how long it took for mean brain volume to increase by 100cm3 over the course of hominin evolution.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Bipedalism

Our anthropoid ancestors of 30–35 million years ago were still tree-dwellers. By about 10 million years ago, the Himalayan mountain range had formed, thrust up in the aftermath of the Indian plate's collision with the Eurasian plate (see Figure 25.16). The climate became drier, and the forests of what are now Africa and Asia contracted. The result was an increased area of savanna (grassland) habitat, with fewer trees. Researchers have hypothesized that as the habitat changed, natural selection may have favored adaptations that made moving over open ground more efficient. Underlying this idea is the fact that while nonhuman apes are superbly adapted for climbing trees, they are less well suited for ground travel. For example, as a chimpanzee walks, it uses four times the amount of energy used by a human.

Although elements of this hypothesis survive, the picture now appears more complex. Even though all fossils of early hominins show indications of bipedalism, none of these hominins lived in savannas. Instead, they

lived in mixed habitats ranging from forests to open woodlands. Furthermore, whatever the selective pressure that led to bipedalism, hominins did not become more bipedal in a simple, linear fashion. *Ardipithecus* had skeletal elements indicating that it could switch to upright walking but also was well suited for climbing trees. Australopiths seem to have had various locomotor styles, and some species spent more time on the ground than others. Only about 1.9 million years ago did hominins begin to walk long distances on two legs. These hominins lived in more arid environments, where bipedal walking requires less energy than walking on all fours.

Tool Use

As you read earlier, the manufacture and use of complex tools are derived behavioral characters of humans. Determining the origin of tool use in hominin evolution is very difficult. Other apes are capable of surprisingly sophisticated tool use. Orangutans, for example, can fashion sticks into probes for retrieving insects from their nests. Chimpanzees are even more adept, using rocks to smash open food and putting leaves on their feet to walk over thorns. It's likely that early hominins were capable of this sort of simple tool use, but finding fossils of modified sticks or leaves that were used as shoes is practically impossible.

The oldest generally accepted evidence of tool use by hominins is 2.5-million-year-old cut marks on animal bones found in Ethiopia. These marks suggest that hominins cut flesh from the bones of animals using stone tools. Interestingly, the hominins whose fossils were found near the site where the bones were discovered had a relatively small brain. If these hominins, which have been named *Australopithecus garhi*, were in fact the creators of the stone tools used on the bones, that would suggest that stone tool use originated before the evolution of large brains in hominins.

HHMI BioInteractive Video

The Tool-Making Animal

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Early *Homo*

The earliest fossils that paleoanthropologists place in our genus, *Homo*, include those of the species *Homo habilis*. These fossils, ranging in age from about 2.4 to 1.6 million years, show clear signs of certain derived hominin characters above the neck. Compared to the australopiths, *H. habilis* had a shorter jaw and a larger brain volume, about 550–750 cm3. Sharp stone tools have also been found with some fossils of *H. habilis* (the name means "handy man").

Fossils from 1.9 to 1.5 million years ago mark a new stage in hominin evolution. A number of paleoanthropologists recognize these fossils as those of a distinct species, *Homo ergaster*. *Homo ergaster* had a substantially larger brain than *H. habilis* (over 900 cm3), as well as long, slender legs with hip joints well adapted for long-distance walking **(Figure 34.50)**. The fingers were relatively short and straight, suggesting that *H. ergaster* did not climb trees like earlier hominins. *Homo ergaster* fossils have been discovered in far more arid environments than earlier hominins and have been associated with more sophisticated stone tools. Its smaller teeth also suggest that *H. ergaster* either ate different foods than australopiths (more meat and less plant material) or prepared some of its food before chewing, perhaps by cooking or mashing the food.

Figure 34.50

Fossil of *Homo ergaster*.

Fossil skeleton of Homo ergaster resembles a modern human skeleton, with skull, ribcage, some arm bones, pelvis, and leg bones visible.

This 1.7-million-year-old fossil from Kenya belongs to a young *Homo ergaster* male. This individual was tall, slender, and fully bipedal, and he had a relatively large brain.

Homo ergaster marks an important shift in the relative sizes of the sexes. In primates, a size difference between males and females is a major component of sexual dimorphism (see Concept 23.4). On average, male gorillas and orangutans weigh about twice as much as females of their species. In *Australopithecus afarensis*, males were 1.5 times as heavy as females. The extent of sexual dimorphism decreased further in early *Homo*, a trend that continues through our own species: Human males weigh only about 1.2 times as much as females.

The reduced sexual dimorphism may offer some clues to the social systems of extinct hominins. In extant primates, extreme sexual dimorphism is associated with intense male-male competition for multiple females. In species that undergo more pair-bonding (including our own), sexual dimorphism is less dramatic. In *H. ergaster*, therefore, males and females may have engaged in more pair-bonding than earlier hominins did.

Fossils now generally recognized as *H. ergaster* once were considered early members of another species, *Homo erectus*, and some paleoanthropologists still hold this position. *Homo erectus* originated in Africa and was the first hominin to migrate out of Africa. The oldest fossils of hominins outside Africa, dating back 1.8 million years, were discovered in the present-day country of Georgia. *Homo erectus* eventually migrated as far as the Indonesian archipelago. Fossil evidence indicates that *H. erectus* became extinct between 200,000 and 70,000 years ago.

Neanderthals

In 1856, miners discovered some mysterious human fossils in a cave in the Neander Valley in Germany. The 40,000-year-old fossils belonged to a thick-boned hominin with a prominent brow. The hominin was named *Homo neanderthalensis* and is commonly called a Neanderthal. Neanderthals were living in Europe by 350,000 years ago and later spread to the Near East, central Asia, and southern Siberia. They had a brain larger than that of present-day humans, buried their dead, and made hunting tools from stone and wood. But despite their adaptations and culture, Neanderthals became extinct at some point between 28,000 and 40,000 years ago.

What is the evolutionary relationship of Neanderthals to *Homo sapiens*? Genetic data indicate that the lineages leading to *H. sapiens* and to Neanderthals diverged about 600,000 years ago **(Figure 34.51)**. This indicates that while Neanderthals and humans share a relatively recent common ancestor, humans did not descend directly from Neanderthals (as was once thought). Neanderthals were, in fact, more closely related to the "Denisovans," a hominin whose fossils have been discovered in Siberia and Tibet, than to humans. (Such studies are possible because scientists have been able to extract DNA from Neanderthal and Denisovan fossils and sequence their genomes. In 2022, Swedish evolutionary geneticist Svante Pääbo was awarded the Nobel Prize in Physiology or Medicine for his research on the genomes of extinct hominins and findings about human evolution.)

Figure 34.51

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    Evolutionary relationships and interbreeding among humans,
Neanderthals, and Denisovans.
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An evolutionary tree.

The first branch begins at a node about 600,000 years ago. The node splits into two branches, branch 1 and 7. Branch 1 labeled as modern humans splits into branch 2 labeled in Africa and branch 3, about 200,000 years ago. Branch 3 splits into branches 4 (labeled in Europe) and branch 5. Branch 5 further splits into branch 6 (labeled in Asia) and branch 7 (labeled in Australia/Pacific Islands). Branch 7 reaches a node about 450,000 years ago and splits into branch 8 (Neanderthals) and branch 9 (Denisovans). There was interbreeding between Neanderthals and non-African modern humans, Neanderthals and Denisovans, Denisovans and Australia/Pacific islands modern humans, and Denisovans and modern humans in Asia The interbreeding is depicted as two headed arrows with hashed lines.

Genomic data and fossil evidence (see Figure 34.52) document multiple examples of interbreeding among humans, Neanderthals, and Denisovans. (Later migrations of *H*. sapiens individuals and mating between human populations have resulted in further distribution of Neanderthal and Denisovan DNA among *H*. sapiens globally.)

Another long-standing question is whether mating occurred between Neanderthals and humans, leading to interspecific gene flow. Some researchers have argued that evidence of gene flow can be found in fossils that show a mixture of human and Neanderthal characteristics. Recent analyses of the DNA sequence of the Neanderthal genome indicated that limited gene flow did occur between the two species. In 2015, the most extensive evidence yet of such gene flow was reported: DNA extracted from the fossil of a human jawbone was found to contain long stretches of Neanderthal DNA **(Figure 34.52a)**. In fact, the amount of Neanderthal DNA in this fossil indicated that this individual's great-great-great-grandparent was a Neanderthal.

Figure 34.52

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Fossil evidence of interbreeding among hominin species.
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Part a. Photo of fossil jawbone shows sturdy structure and the remaining molars look large.
Part b. Photo of a bone fragment.

(a) This jawbone belonged to a human who lived 40,000 years ago and had a relatively recent Neanderthal ancestor. **(b)** This bone fragment came from a female whose parents were a Neanderthal female and a Denisovan male.

Other recent genomic studies have shown that gene flow also occurred between Neanderthals and the Denisovans. In 2018, for example, DNA analyses revealed that the 90,000-year-old bone fragment shown in **Figure 34.52b** came from a female whose parents were a Neanderthal female and a Denisovan male—a remarkable discovery documenting the direct offspring of two different species of hominins. Genomic analyses also indicate that gene flow occurred between the Denisovans and our own species, *H. sapiens*.

Homo sapiens

Evidence from fossils, archaeology, and DNA studies has improved our understanding about how our own species, *Homo sapiens*, emerged and spread around the world.

Fossil evidence indicates that the ancestors of humans originated in Africa. Older species gave rise to later species, ultimately including *H. sapiens*. A 2017 study indicates that by 315,000 years ago, some hominin

populations in Morocco had facial features like those of our species, while the back of the skull remained elongated, as in earlier species. The oldest known fossils of our species include specimens from Ethiopia that are 195,000 and 160,000 years old **(Figure 34.53)**. These early humans had less pronounced browridges than those found in *H. erectus* and Neanderthals, and they were more slender than other recent hominins. The Ethiopian fossils support inferences about the origin of humans from molecular evidence. DNA analyses show that Europeans and Asians share a relatively recent common ancestor and that many African lineages branched off more basal positions on the human family tree. These findings suggest that all living humans have ancestors that originated as *H. sapiens* in Africa.

Figure 34.53

A 160,000-year-old-fossil of *Homo sapiens*.

Homo sapiens skull is mostly intact, with a few gaps. Eye sockets and upper jaw prominent.

The oldest fossils of *H. sapiens* outside Africa are from the Middle East and date back about 180,000 years. Fossil evidence and genetic analyses suggest that humans spread beyond Africa in one or more waves, first into Asia and then to Europe and Australia. The date of the first arrival of humans in the Americas is uncertain. Recent archaeological discoveries—including 22,000-year-old human footprints found at White Sands National Park in New Mexico—indicate that humans had reached the Americas by about 25,000 years ago. However, other scientists are skeptical of these findings and put that date at 16,000 years ago.

New findings continually update our understanding of the human evolutionary lineage. For example, in 2015, the human family gained a new member, *Homo naledi*. The structure of its foot indicates that *H. naledi* was fully bipedal, and the shape of its hand suggests that *H. naledi* had fine motor skills **(Figure 34.54)**, as in *H. sapiens*, Neanderthals, and other species that used tools extensively. But *H. naledi* also had a small brain, a broadly flared upper pelvis, and other features that have led researchers to conclude that it was an early member of our genus.

Figure 34.54

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    Fossils of hand bones and foot bones (top and side views) of
*Homo naledi.*
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Photo on left shows long slender finger bones. Photo on right shows delicate slender foot bones.

As an early member of our genus, it is likely that *H. naledi* originated more than 1.5 million years ago—yet initial estimates of the age of the *H. naledi* fossils ranged from 3 million years old to just 100,000 years old.

Scientists didn't know how old the fossils are because they were found on the floor of a deep cave, not encased in rocks that could be dated using radioactive isotopes. In 2017, however, teeth from a newly discovered *H. naledi* skull were dated using radioactive isotopes and found to be 300,000 years old. Overall, fossil evidence suggests that *H. naledi* arose 1–2 million years ago (like other early members of our genus) and then persisted almost until the present.

About ten years before the discovery of *H. naledi*, researchers reported another astonishing find: skeletal remains of adult hominins dating from 100,000 to 60,000 years ago and representing a previously unknown species, *Homo floresiensis*. First discovered in a limestone cave on the Indonesian island of Flores, the individuals were much shorter and had a much smaller brain volume than *H. sapiens*—more similar, in fact, to an australopith. The researchers who discovered these fossils argue that certain features of the skeletons, such as the shape of the teeth and the thickness and proportions of the skull, suggest that *H. floresiensis* descended from the larger *H. erectus*. Not convinced, some researchers have argued that the fossils represent small *H. sapiens* individuals with a disorder such as Down syndrome or microcephaly (a condition in which a person has a deformed, miniature brain).

Over the past few years, most studies have supported the designation of *H. floresiensis* as a new hominin. One such study found that the wrist bones of the Flores fossils are similar in shape to those of nonhuman apes and early hominins, but different from those of Neanderthals and *H. sapiens*. These researchers concluded that the Flores fossils represent a species whose lineage branched off before the origin of the clade that includes Neanderthals and humans. A different study comparing the foot bones of the Flores fossils with those of other hominins also concluded that *H. floresiensis* arose before *H. sapiens*. Finally, in 2016, *H. floresiensis*-like fossils dating to 700,000 years ago were discovered at a second location on Flores—again indicating that the *H. floresiensis* lineage branched off long before the origin of *H. sapiens*. Open questions that may yet be answered by new discoveries include how *H. floresiensis* originated and whether it survived long enough to encounter *H. sapiens*, which was living in Indonesia by 50,000 years ago.

The rapid expansion of our species may have been spurred by changes in human cognition as *H. sapiens* evolved in Africa. Evidence of sophisticated thought in *H. sapiens* includes the recent discovery in South Africa of 77,000-year-old art—geometric markings made on pieces of ochre **(Figure 34.55)**. Similarly, archaeologists working in southern and eastern Africa recently found 75,000-year-old ostrich eggs and snail shells with holes neatly drilled through them. By 30,000 years ago, humans were producing spectacular cave paintings.

Figure 34.55

Art, a human hallmark.

Ochre with engraved lines criss crossing surface.

The engravings on this 77,000-year-old piece of ochre, discovered in South Africa's Blombos Cave, are among the earliest signs of symbolic thought in humans.

While these developments can help us understand the spread of *H. sapiens*, it is not clear whether they played a role in the extinction of other hominins. Neanderthals, for example, also made complex tools and showed a capacity for symbolic thought. As a result, while some scientists have suggested that Neanderthals were driven to extinction by competition with *H. sapiens*, others question that idea.

Our discussion of humans brings this unit on biological diversity to an end. But keep in mind that our sequence of topics isn't meant to imply that life consists of a ladder leading from lowly microorganisms to lofty humanity. Biological diversity is the product of branching phylogeny, not ladderlike "progress." The fact that there are almost as many species of ray-finned fishes alive today as in all other vertebrate groups combined shows that our finned relatives are not outmoded underachievers that failed to leave the water. The tetrapods—amphibians, reptiles, and mammals—are derived from one lineage of lobe-finned vertebrates. As tetrapods diversified on land, fishes continued their branching evolution in the greatest portion of the biosphere's volume. Similarly, the ubiquity of diverse prokaryotes throughout the biosphere today is a reminder of the enduring ability of these relatively simple organisms to keep up with the times through adaptive evolution. Biology exalts life's diversity, past and present.